SURJECTIONS OF MODULES INDUCE CANONICAL CLOSED IMMERSIONS OF QUOT FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 924-930 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 932-947, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_moduli.json (pb=777, stmt 924-930, proof 932-947).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Source typo 'sequecnce' retained verbatim; harmless.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 083M: sources/stacks-moduli/pinned-original-source.tex, source lines 947–947. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to B$ be as in the introduction to this section. Let $\mathcal{F} \to \mathcal{G}$ be a surjection of quasi-coherent $\mathcal{O}_X$-modules. Then there is a canonical closed immersion $\mathrm{Quot}_{\mathcal{G}/X/B} \to \mathrm{Quot}_{\mathcal{F}/X/B}$.*

Proof. Let $\mathcal{K} = \mathrm{Ker}(\mathcal{F} \to \mathcal{G})$. By right exactness of pullbacks we find that $\mathcal{K}_T \to \mathcal{F}_T \to \mathcal{G}_T \to 0$ is an exact sequecnce for all schemes T over B. In particular, a quotient of $\mathcal{G}_T$ determines a quotient of $\mathcal{F}_T$ and we obtain our transformation of functors $\mathrm{Quot}_{\mathcal{G}/X/B} \to \mathrm{Quot}_{\mathcal{F}/X/B}$. This transformation is a closed immersion by Flatness on Spaces, Lemma 083M. Namely, given an element $\mathcal{F}_T \to \mathcal{Q}$ of $\mathrm{Quot}_{\mathcal{F}/X/B}(T)$, then we see that the pull back to T'/T is in the image of the transformation if and only if $\mathcal{K}_{T'} \to \mathcal{Q}_{T'}$ is zero. □

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.